VIGNESH K

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SUMMARY

AI and Machine Learning and Full Stack (React + FastAPI) developer with experience in building scalable applications, predictive models and deploying AI-powered solutions.

EDUCATION

B.E. Computer Science and Engineering (AI & ML) 2023 – Present
KPR Institute of Engineering and Technology, Coimbatore
CGPA: 8.7 — Honors GPA: 9.0
Jai Saradha Matric Higher Secondary School
Class XII: 91.5% — Class X: 91.5%

TECHNICAL SKILLS

Programming: C, Java, Python, SQL
AI/LLM Frameworks: CrewAI, LangChain, RAG
Machine Learning: Scikit-learn, Transformers, LSTM, Isolation Forest, Decision Trees, SVM, Random Forest, YOLOv8, OpenCV
Web: FastAPI, Flask, HTML, CSS, JavaScript, React, Bootstrap, Tailwind CSS, Framer Motion
Databases: MySQL, MongoDB, PostgreSQL
Libraries: Pandas, NumPy, Matplotlib

EXPERIENCE

AI and ML — Payoda Technology Feb 2026 - Present
- Developed an AI-powered testing app using LLMs and CrewAI for automated test case generation.
- Built backend services for real-time test execution status tracking and automated email/WhatsApp notifications.

Machine Learning Intern — Synovers Technologies Jan 2025 – Feb 2025
- Built an osteoporosis prediction system using Decision Tree Classifier.
- Performed preprocessing, feature engineering and model evaluation.

PROJECTS

AI-Powered Government Scheme Eligibility and Recommendation System Present
- Built an AI-based government scheme finder using Python, React, FastAPI, RAG, LLMs, and LangChain for personalized scheme recommendations.
- Developed a retrieval pipeline using Vector Database to provide accurate scheme information through AI assistant.

Healthlens AI - Present
- Developed a RAG pipeline using ChromaDB for semantic search and AI-powered medical report analysis.
- Built a user-friendly web interface for uploading reports and generating context-aware responses using an LLM.

AI-based Smart Hostel Anomaly Detection system - Jan 2026 - Mar 2026
- Built an AI-based hostel anomaly detection system for real-time identification of abnormal activities with automated alerts. Implemented real-time alerting and evidence capture.

GYM Fit Track App Dec 2025
- Built a fitness tracking app using React.js and FastAPI.
- Integrated a RAG-based AI assistant for gym document retrieval and context-aware user query responses.

Osteoporosis Risk Prediction System - March 2024
- Developed Logistic Regression, Random Forest and SVM models.
- Optimized features to improve predictive performance. Delivered an interactive clinical web interface.

PUBLICATIONS

Generative AI for Pesonalized Energy Consumption Behavior Analysis — IGI Global Dec 2025
- Uses Generative AI to predict and optimize personalized energy usage.
- Enhances resilient, ethical energy systems with human behavior insights.

Osteoporosis Prediction Using Machine Learning:XGBoost Approach — IEEE Xplore Aug 2025
- Developed an osteoporosis prediction model using XGBoost, achieving 92 percent accuracy and 98 percent precision on real-world clinical datasets.Cost-effective, scalable ML solution enabling early disease detection.

Enhanced Brain Tumor Detection using CNNs — IGI Global May 2025
- Worked on tumor segmentation and classification using deep learning architectures.
- Collaborated with a research team to improve detection accuracy.

CERTIFICATIONS

- AlgoUniversity Tech Fellowship 2025 — AlgoUniversity
- Student Jury Mobility 2.0 24hr Hackathon — IEEE VTS